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spdfiv: Defactored Instrumental Variables Estimation and Heterogeneity Diagnostics for Spatial Dynamic Panels with Interactive Effects in Python

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Abstract

Spatial dynamic panel data models with unobserved common factors can be estimated by the two-stage instrumental variables (2SIV) estimator of Cui, Sarafidis, and Yamagata (2023) and, with heterogeneous slopes, by the mean-group IV estimator of Chen, Cui, Sarafidis, and Yamagata (2025). This article introduces **spdfiv**, an open-source Python implementation of both estimators, to our knowledge the first outside Stata. The package reproduces the published Stata results of Kripfganz and Sarafidis (2025) in every printed digit; the comparison uncovered five implementation details that are not visible from the published equations, which we document. The package also matches the weights matrix to the panel by unit label, computes impacts exactly through a Schur decomposition with Delta-method or simulation inference, provides diagnostics that show which units drive a mean-group estimate, and offers an experimental treatment of unbalanced panels. Monte Carlo experiments document each feature. Applied to the bank data of Kripfganz and Sarafidis (2025), the diagnostics show that extreme bank-level estimates explain most of the gap between the mean-group and pooled estimates of some slopes, but not the disappearance of the spatial spillover.

Keywords: panel data, spatial econometrics, interactive effects, common factors, instrumental variables, mean-group estimation, outliers, Python.

1. Introduction

Panel data that follow many units over many periods allow researchers to separate three sources of dependence that are confounded in a single cross section or time series: state dependence, captured by lags of the outcome; spatial or network interactions, captured by spatially weighted averages of the outcomes of other units; and common shocks, such as

business cycles, financial crises, or regulatory changes, that hit all units with heterogeneous intensities. The last source is commonly modelled by a factor structure $\boldsymbol{\gamma}_i^\top \boldsymbol{f}_t$, also known as interactive effects (Pesaran 2006; Bai 2003).

Combining all three features is demanding. Quasi-maximum likelihood estimators (Yu, de Jong, and Lee 2008; Shi and Lee 2017; Bai and Li 2021) are non-linear, require bias correction for incidental parameters, and do not accommodate covariates that are endogenous with respect to the idiosyncratic error. Cui, Sarafidis, and Yamagata (2023) proposed a two-stage instrumental variables (2SIV) estimator that avoids all three problems: it is linear in the parameters, free from asymptotic bias, and allows external instruments. Chen, Cui, Sarafidis, and Yamagata (2025) extended the approach to heterogeneous slopes with a mean-group IV (MGIV) estimator. Kripfganz and Sarafidis (2025) implemented both estimators in the Stata command `spxtivdfreg` and noted that the approach had not been implemented in any other statistical software.

This article introduces **spdfiv**, a Python (Python Software Foundation 2024) package that fills this gap. Its dependencies are limited to **NumPy** (Harris, Millman, van der Walt *et al.* 2020), **SciPy** (Virtanen, Gommers, Oliphant *et al.* 2020), and **pandas** (McKinney 2010), so it is available to every user of the scientific Python stack at no cost. Spatial econometric software in Python, notably the **spreg** package of the **PySAL** ecosystem (Rey and Anselin 2010; Rey, Anselin, Amaral, Arribas-Bel, Cortes, Gaboardi, Kang, Knaap, Li, Lumnitz, Oshan, Shao, and Wolf 2022), does not provide estimators with interactive effects; neither do **splm** (Millo and Piras 2012) in R or **xsmle** (Belotti, Hughes, and Mortari 2017) in Stata.

A translation of an existing command would be of limited interest on its own. The package therefore also addresses four limitations that Kripfganz and Sarafidis (2025) acknowledge in their own discussion.

1. *Alignment of the spatial weights matrix.* `spxtivdfreg` does not check that the rows of the weights matrix are ordered like the units in the data, leaving this to the user. **spdfiv** matches the matrix to the panel by unit label and refuses unlabelled input unless the user explicitly accepts positional matching.

2. *Inference on impacts.* Standard errors of direct, indirect, and total impacts are available only through the Delta method. Long-run multipliers such as $\beta/(1-\rho-\psi)$ are highly non-linear when the process is persistent. **spdfiv** adds Krinsky–Robb simulation (Krinsky and Robb 1986), the approach of LeSage and Pace (2009), and computes impacts through a Schur decomposition that remains exact when the weights matrix is not diagonalizable.

3. *Outliers in mean-group estimation.* In the application of Kripfganz and Sarafidis (2025), the MG estimates differ from the pooled ones by an order of magnitude for some coefficients, and the authors conjecture that a few outlying units are responsible, leaving the question for future work. **spdfiv** provides tools to answer it: influence statistics for each unit, outlier flags, and robust summaries of the distribution of unit-specific estimates with bootstrap inference.

4. *Unbalanced panels.* `spxtivdfreg` requires a balanced panel. **spdfiv** accepts panels with staggered entry or sporadic gaps through an explicit, experimental option, documented together with its limitations and with simulation evidence of its performance.

The package is validated in three ways. A test suite re-implements the 2SIV estimator literally from its defining equations and checks that the package reproduces estimates, variance matrix, and J statistic to machine precision; Monte Carlo experiments document finite-sample performance; and, on the bank data of Kripfganz and Sarafidis (2025), the package reproduces their published 2SIV and mean-group estimates, standard errors, J statistic, and long-run impacts in every printed digit. This cross-software check uncovered five implementation details of `spxtivdfreg` that are not visible from the equations of the article but change the results substantially; they are the defaults of **spdfiv** and are documented in Section 6.

Section 2 reviews the estimators and describes the extensions. Section 3 presents the package, Section 4 illustrates its use, Section 5 reports Monte Carlo evidence, Section 6 applies the package to the bank data, Section 7 reports computing times, and Section 8 concludes.

2. Models and methods

2.1. Model

For units $i = 1, \dots, N$ and periods $t = 1, \dots, T$, consider

$$y_{it} = \sum_{\tau=0}^{S_\psi} \psi_\tau \sum_{j=1}^{N} w_{ij} y_{j,t-\tau} + \sum_{\tau=1}^{S_\rho} \rho_\tau y_{i,t-\tau} + \boldsymbol{x}_{it}^\top \boldsymbol{\beta} + \sum_{j=1}^{N} w_{ij} \boldsymbol{x}_{jt}^\top \boldsymbol{\delta} + \alpha_i + \boldsymbol{\gamma}_i^\top \boldsymbol{f}_t + \varepsilon_{it}, \tag{1}$$

where w_{ij} are the elements of a known $N \times N$ spatial weights matrix $\boldsymbol{W}$ with zero diagonal, α_i are unit effects, $\boldsymbol{f}_t$ are r_y unobserved factors with loadings $\boldsymbol{\gamma}_i$, and ε_{it} is an idiosyncratic error. The covariates follow

$$\boldsymbol{x}_{it} = \boldsymbol{\Gamma}_i^\top \boldsymbol{f}_{x,t} + \boldsymbol{v}_{it}, \tag{2}$$

so that they are correlated with the factor component of the error. This is model (23) of Kripfganz and Sarafidis (2025); with $S_\psi = 0$, $S_\rho = 1$ and $\boldsymbol{\delta} = \boldsymbol{0}$ it reduces to the model of Cui *et al.* (2023). Stacking the T observations of unit i gives $\boldsymbol{y}_i = \boldsymbol{C}_i \boldsymbol{\theta} + \boldsymbol{u}_i$, where $\boldsymbol{C}_i$ collects the regressors and $\boldsymbol{u}_i = \boldsymbol{F}\boldsymbol{\gamma}_i + \boldsymbol{\varepsilon}_i$.

2.2. Two-stage IV estimation

Let $\hat{\boldsymbol{F}}_{x,-\tau}$ be $\sqrt{T}$ times the eigenvectors of the r_x largest eigenvalues of $(NT)^{-1} \sum_i \boldsymbol{X}_{i,-\tau} \boldsymbol{X}_{i,-\tau}^\top$, where $\boldsymbol{X}_{i,-\tau}$ holds the instrumenting covariates lagged τ periods, and let $\boldsymbol{M}_{\hat{F}_{x,-\tau}}$ be the projection that removes them. The instrument matrix is

$$\hat{\boldsymbol{Z}}_i = \left(\boldsymbol{M}_{\hat{F}_{x,-\tau}} \boldsymbol{X}_{i,-\tau},\ \textstyle\sum_j w_{ij} \boldsymbol{M}_{\hat{F}_{x,-\tau}} \boldsymbol{X}_{j,-\tau} \right)_{\tau=0,\dots,L}. \tag{3}$$

The first-stage estimator is the IV estimator $\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{A}}^\top \hat{\boldsymbol{B}}^{-1} \hat{\boldsymbol{A}})^{-1} \hat{\boldsymbol{A}}^\top \hat{\boldsymbol{B}}^{-1} \hat{\boldsymbol{c}}$ with $\hat{\boldsymbol{A}} = \sum_i \hat{\boldsymbol{Z}}_i^\top \boldsymbol{C}_i$, $\hat{\boldsymbol{B}} = \sum_i \hat{\boldsymbol{Z}}_i^\top \hat{\boldsymbol{Z}}_i$, and $\hat{\boldsymbol{c}} = \sum_i \hat{\boldsymbol{Z}}_i^\top \boldsymbol{y}_i$. In the second stage the factors $\hat{\boldsymbol{F}}_y$ are estimated from the first-stage residuals $\hat{\boldsymbol{u}}_i$, projected out by $\boldsymbol{M}_{\hat{F}_y}$, and

$$\tilde{\boldsymbol{\theta}} = (\tilde{\boldsymbol{A}}^\top \tilde{\boldsymbol{B}}^{-1} \tilde{\boldsymbol{A}})^{-1} \tilde{\boldsymbol{A}}^\top \tilde{\boldsymbol{B}}^{-1} \tilde{\boldsymbol{c}}, \quad \tilde{\boldsymbol{A}} = \sum_i \hat{\boldsymbol{Z}}_i^\top \boldsymbol{M}_{\hat{F}_y} \boldsymbol{C}_i,\ \tilde{\boldsymbol{B}} = \sum_i \hat{\boldsymbol{Z}}_i^\top \boldsymbol{M}_{\hat{F}_y} \hat{\boldsymbol{u}}_i \hat{\boldsymbol{u}}_i^\top \boldsymbol{M}_{\hat{F}_y} \hat{\boldsymbol{Z}}_i. \tag{4}$$

Cui *et al.* (2023) show that $\tilde{\boldsymbol{\theta}}$ is consistent, asymptotically normal, and correctly centred as $N, T \to \infty$ with $N/T \to c \in (0, \infty)$. The package uses the robust variance

$$\hat{\boldsymbol{V}} = (\tilde{\boldsymbol{A}}^\top \tilde{\boldsymbol{B}}^{-1} \tilde{\boldsymbol{A}})^{-1} \tilde{\boldsymbol{A}}^\top \tilde{\boldsymbol{B}}^{-1} \hat{\boldsymbol{\Omega}} \tilde{\boldsymbol{B}}^{-1} \tilde{\boldsymbol{A}} (\tilde{\boldsymbol{A}}^\top \tilde{\boldsymbol{B}}^{-1} \tilde{\boldsymbol{A}})^{-1}, \qquad \hat{\boldsymbol{\Omega}} = \sum_i \hat{\boldsymbol{Z}}_i^\top \tilde{\boldsymbol{e}}_i \tilde{\boldsymbol{e}}_i^\top \hat{\boldsymbol{Z}}_i, \tag{5}$$

which allows heteroskedasticity and serial correlation within units. By default $\hat{\boldsymbol{\Omega}}$ is evaluated at the first-stage residuals, $\hat{\boldsymbol{\Omega}} = \tilde{\boldsymbol{B}}$, as in Equation (11) of Kripfganz and Sarafidis (2025) and in `spxtivdfreg`, so that $\hat{\boldsymbol{V}} = (\tilde{\boldsymbol{A}}^\top \tilde{\boldsymbol{B}}^{-1} \tilde{\boldsymbol{A}})^{-1}$; with `omega="second"` it is evaluated at the second-stage residuals $\tilde{\boldsymbol{e}}_i = \boldsymbol{M}_{\hat{F}_y}(\boldsymbol{y}_i - \boldsymbol{C}_i \tilde{\boldsymbol{\theta}})$. Both choices are consistent. The package reports the overidentification statistic $J = \bar{\boldsymbol{g}}^\top \hat{\boldsymbol{\Omega}}^{-1} \bar{\boldsymbol{g}}$ with $\bar{\boldsymbol{g}} = \sum_i \hat{\boldsymbol{Z}}_i^\top \tilde{\boldsymbol{e}}_i$, asymptotically χ^2 with as many degrees of freedom as there are overidentifying restrictions. The numbers of factors are chosen by the eigenvalue-ratio criterion of Ahn and Horenstein (2013), which selects between one and `factmax` factors; with `zero_factors=True` their mock eigenvalue is added so that zero factors can be selected. Unit effects are removed by a within transformation of all variables, including the instruments, before the factors are extracted.

2.3. Mean-group IV estimation

When $\boldsymbol{\theta}_i = \boldsymbol{\theta} + \boldsymbol{\eta}_i$ with $\boldsymbol{\eta}_i$ independent across units, Chen *et al.* (2025) estimate each $\boldsymbol{\theta}_i$ by IV using the instruments (3), where the lagged blocks are additionally projected on the contemporaneous factors ("double defactoring"), and average: $\hat{\boldsymbol{\theta}}_{\mathrm{MG}} = N^{-1} \sum_i \hat{\boldsymbol{\theta}}_i$ with variance estimator $\hat{\boldsymbol{\Sigma}}_\eta / N$, where $\hat{\boldsymbol{\Sigma}}_\eta = (N-1)^{-1} \sum_i (\hat{\boldsymbol{\theta}}_i - \hat{\boldsymbol{\theta}}_{\mathrm{MG}})(\hat{\boldsymbol{\theta}}_i - \hat{\boldsymbol{\theta}}_{\mathrm{MG}})^\top$ (see also Pesaran and Smith 1995).

2.4. Impacts and their inference

For covariate ℓ the matrix of long-run partial derivatives is

$$\boldsymbol{S}_\ell = [a \boldsymbol{I}_N - b \boldsymbol{W}]^{-1} (\beta_\ell \boldsymbol{I}_N + \delta_\ell \boldsymbol{W}), \qquad a = 1 - \textstyle\sum_\tau \rho_\tau, \quad b = \sum_\tau \psi_\tau, \tag{6}$$

and the short-run matrix sets $a = 1$ and $b = \psi_0$. The average direct impact is $\mathrm{tr}(\boldsymbol{S}_\ell)/N$, the average total impact is $\boldsymbol{1}^\top \boldsymbol{S}_\ell \boldsymbol{1}/N$, and the average indirect impact is their difference (LeSage and Pace 2009; Debarsy, Ertur, and LeSage 2012).

Computation. Let $\boldsymbol{W} = \boldsymbol{Q} \boldsymbol{R} \boldsymbol{Q}^*$ be the complex Schur decomposition, with $\boldsymbol{Q}$ unitary and $\boldsymbol{R}$ upper triangular with the eigenvalues $\lambda_1, \ldots, \lambda_N$ on its diagonal. Because $\boldsymbol{S}_\ell$ is a rational function of $\boldsymbol{W}$, its trace equals $\sum_j (\beta_\ell + \delta_\ell \lambda_j)/(a - b\lambda_j)$ *whether or not* $\boldsymbol{W}$ *is diagonalizable*, and $\boldsymbol{1}^\top \boldsymbol{S}_\ell \boldsymbol{1} = (\boldsymbol{1}^\top \boldsymbol{Q})(a\boldsymbol{I} - b\boldsymbol{R})^{-1} \boldsymbol{Q}^* (\beta_\ell \boldsymbol{1} + \delta_\ell \boldsymbol{W} \boldsymbol{1})$ requires a single triangular solve. When all row sums of $\boldsymbol{W}$ equal s, as after row standardization, the total impact reduces to $(\beta_\ell + \delta_\ell s)/(a - bs)$. After one $O(N^3)$ decomposition, each evaluation costs $O(N)$ for direct impacts and at most $O(N^2)$ for total impacts, which makes simulation-based inference cheap. The Schur form matters in practice: asymmetric k-nearest-neighbour matrices, common in applied work, are often numerically close to defective, and in our experiments the eigenvector matrices of such matrices had condition numbers above 10^{11}, which makes formulas based on $\boldsymbol{W} = \boldsymbol{V} \boldsymbol{\Lambda} \boldsymbol{V}^{-1}$ unreliable.

Inference. Impacts are smooth functions $h(\boldsymbol{\theta})$ of the coefficients. Besides the Delta method, **spdfiv** offers Krinsky–Robb simulation: draws $\boldsymbol{\theta}^{(r)} \sim N(\tilde{\boldsymbol{\theta}}, \hat{\boldsymbol{V}})$ are mapped into $h(\boldsymbol{\theta}^{(r)})$, and percentile intervals are reported. Draws that violate the stability conditions $\psi_0\omega < 1$ and $a - b\omega > 0$, with ω the spectral radius of $\boldsymbol{W}$, are discarded and their share is reported. The two methods share the same first-order justification; the simulation intervals are not forced to be symmetric, which matters when $a - b\omega$ is close to zero and h is strongly convex.

2.5. Diagnostics for heterogeneous slopes

Under the random-coefficient assumption the estimates $\hat{\boldsymbol{\theta}}_i$ are, up to estimation error, independent draws from the distribution of $\boldsymbol{\theta}_i$, and the MG estimator is the sample mean of that distribution. A sample mean is sensitive to a few extreme draws, which may reflect data errors, structural breaks, or units with nearly uninformative instruments. **spdfiv** offers three tools.

1. *Influence.* Deleting unit i changes the MG estimate by $\hat{\boldsymbol{\theta}}_{\mathrm{MG}} - \hat{\boldsymbol{\theta}}_{\mathrm{MG}(-i)} = (\hat{\boldsymbol{\theta}}_i - \hat{\boldsymbol{\theta}}_{\mathrm{MG}})/(N-1)$. The package reports this change in units of the MG standard error.

2. *Outlier flags.* The modified z-score $0.6745(\hat{\theta}_{i\ell} - \mathrm{med}_\ell)/\mathrm{MAD}_\ell$ of Iglewicz and Hoaglin (1993), with the median and the median absolute deviation computed across units, flags estimates with $|z| > 3.5$.

3. *Robust summaries.* Coefficient-wise trimmed means and medians (Huber and Ronchetti 2009), with standard errors and percentile intervals from a nonparametric bootstrap that resamples units (Efron and Tibshirani 1993); resampling units is justified by the independence of $\boldsymbol{\eta}_i$ across units.

The influence table also reports, for each unit, the condition number of $\hat{\boldsymbol{Z}}_i^\top \hat{\boldsymbol{Z}}_i$; very large values mark units whose instruments are (nearly) collinear, for example because a covariate does not vary over time for that unit, so that their estimates depend on numerical tolerances rather than on the data.

Two caveats are stated in the documentation and bear repeating. The trimmed mean and the median estimate the centre of the distribution of $\boldsymbol{\theta}_i$, which equals the mean $\boldsymbol{\theta}$ only if the distribution is symmetric. And the tools are diagnostics: they show which units drive an estimate, but whether those units should be excluded is a substantive decision for the researcher.

2.6. Unbalanced panels (experimental)

Two departures from balance are supported. With *staggered* entry or exit, each unit is observed over one contiguous spell; with *sporadic* gaps, interior observations may also be missing. In both cases three ingredients change.

1. The spatial lag in period t uses the neighbours available in that period. By default the remaining weights in each row are rescaled to keep the original row sum, so that a balanced panel is analysed with exactly $\boldsymbol{W}$; the alternative, dropping absent neighbours without rescaling, is available because, as Kelejian and Prucha (2010) stress, normalization is a modelling choice.

2. Factors are extracted from the $T \times T$ matrix whose (s, t) element averages over the units observed in both periods, the "all-purpose" estimator of Xiong and Pelger (2023); for a balanced panel it coincides with the usual matrix.

3. The projection $\boldsymbol{M}_{\hat{F}}$ for unit i uses only the rows of $\hat{\boldsymbol{F}}$ for the periods in which unit i is observed.

The theory of Cui *et al.* (2023) and Chen *et al.* (2025) assumes a balanced panel, and extending it is beyond the scope of a software article. The option is therefore labelled experimental, triggers a warning, and is evaluated by simulation in Section 5. The two cases differ in substance. Staggered entry is consistent with a model in which a unit does not exist before it enters, so that the period-specific weights are the correct ones. Sporadic gaps in units that do exist are a missing-data problem: the spatial lag computed from observed neighbours then differs from the true one, which can bias the spatial coefficient (cf. Wang and Lee 2013).

3. The spdfiv package

3.1. Design

The package is written in pure `Python`. All computations are vectorized with **NumPy** or reduce to per-unit linear algebra on small matrices, so the cost grows linearly in N apart from the eigen decompositions of $T \times T$ matrices for the factors and one Schur decomposition of $\boldsymbol{W}$ for impacts. The main entry point is `fit()`; it returns a results object with methods for impacts and heterogeneity diagnostics. Table 1 maps its arguments to the options of `spxtivdfreg`, to ease migration and cross-checks.

3.2. Results

The object returned by `fit()` stores the coefficients (`params`), their variance matrix (`cov`), standard errors (`bse`), z statistics (`zvalues`), p values (`pvalues`), confidence intervals (`conf_int()`), the numbers of factors (`factors_x`, one per instrument lag block, and `factors_u`), the J statistic with its degrees of freedom and p value, and the decomposition of the error variance into factor and idiosyncratic parts. For the MG estimator, `unit_params` and `unit_se` hold the unit-specific estimates. The methods `impacts()`, `mg_influence()`, and `mg_robust()` implement Sections 2.4 and 2.5, and `summary()` prints a coefficient table.

3.3. Alignment of the weights matrix

The weights matrix is passed as a **pandas** `DataFrame` whose index and columns are unit identifiers, or as an array together with the identifiers (`W_ids`). The package checks that the labels on both axes agree, that they coincide with the units in the data, that there are no missing values, and that the diagonal is zero; it then reorders the matrix to the internal ordering of units. An unlabelled array is rejected unless `assume_ordered=True`. The helper `read_weights()` reads a labelled matrix from a CSV or Excel file.

spdfiv argument	`spxtivdfreg` option	Meaning
`W=` (labelled)	`spmatrix()`	spatial weights matrix
`splag=True`	`splag`	spatial lag of y
`tlags=`p	`tlags(`p`)`	time lags of y
`sptlags=`q	`sptlags(`q`)`	spatial time lags of y
`spx=[...]`	`spindepvars()`	spatial lags of covariates
`iv=[...]`	`iv(`*varlist*`)`	exogenous instrumenting variables
`iv_lags=`L	`iv(..., lags(`L`))`	lags of instruments
`iv_splags=True`	`iv(..., splags)`	spatial lags of instruments
`factmax=`k	`factmax(`k`)`	maximum number of factors
`eigratio=False`	`noeigratio`	use exactly `factmax` factors
`zero_factors=True`	—	allow zero factors (mock eigenvalue)
`common_rank=True`	(always)	same number of factors for all lag blocks
`omega="first"`	(always)	$\hat{\boldsymbol{\Omega}}$ from first-stage residuals
`std=True`	`std`	standardize within periods before factor extraction
`absorb=True`	`absorb(`*id*`)`	unit fixed effects
`estimator="fstage"`	`fstage`	first-stage IV
`estimator="mg"`	`mg`	mean-group IV
`doubledefact=`	`doubledefact`	double defactoring of lagged instruments
`unbalanced=`	—	staggered or sporadic missingness (experimental)
`renormalize=`	—	row rescaling with absent neighbours

Table 1: Arguments of `fit()` and corresponding options of the Stata command `spxtivdfreg`.

3.4. Validation

The test suite, run with `python -m unittest`, contains a literal implementation of equations (5)–(12) of Kripfganz and Sarafidis (2025), for both choices of $\hat{\boldsymbol{\Omega}}$, written with explicit $T \times T$ projection matrices and independent of the package code. The package reproduces its first- and second-stage estimates to an absolute tolerance of 10^{-10}, its variance matrix to a relative tolerance of 10^{-8}, and its J statistic to eight decimal places. With no factors and no lags the first-stage estimator coincides with two-stage least squares, which is also tested. Further tests check that permuting a labelled weights matrix leaves the estimates unchanged, that misaligned or invalid matrices are rejected, that the Schur-based impacts agree with dense matrix inversion for both row-standardized and spectrally normalized matrices, that the balanced-panel estimates are unchanged by the unbalanced option, and that the influence statistic equals the leave-one-out change in the MG estimate.

4. Illustration

This section uses simulated data so that the true parameters are known and the output can be reproduced exactly. The data-generating process is model (1) with $\psi_0 = 0.3$, $\rho_1 = 0.4$, $\boldsymbol{\beta} = (1, 0.5)^\top$, two AR(1) factors that enter both the error and the covariates, loadings of the covariates correlated with those of the error, heteroskedastic errors, unit effects, and a row-standardized six-nearest-neighbour weights matrix; it is available as `simulate_panel()`.

```
>>> import pandas as pd
>>> import spdfiv as sd
```

```
>>> data, W, truth = sd.simulate_panel(N=300, T=40, psi=0.3, rho=0.4,
...                                    beta=(1.0, 0.5), seed=2025)
>>> print(data.head(3)); print(W.shape, W.index[:5].tolist())

   id  time        y      x1      x2
0   1     1 -18.3217 -6.0533 -1.5710
1   1     2 -11.4466 -1.3434  0.4856
2   1     3  -8.7663 -2.4066  0.5138
(300, 300) [1, 2, 3, 4, 5]
```

The data are in long format; `W` is a labelled `DataFrame`. The 2SIV estimator with a spatial lag, one time lag, and instruments formed from the defactored covariates, their first lags, and the spatial lags of both, is obtained with

```
>>> res = sd.fit(data, y="y", x=["x1", "x2"], unit="id", time="time", W=W,
...              splag=True, tlags=1, iv_lags=1, iv_splags=True)
>>> print(res.summary())

Defactored instrumental variables estimation
Number of obs               =      11700   Number of units     =      300
Number of instruments       =          8   Obs per unit (min/avg/max)= 39/39.0/39
Factors in instruments      =        2/2   Factors in u        =        2
Second-stage estimator (model with homogeneous slope coefficients)
------------------------------------------------------------------------------
                     Coef.   Std. err.        z   P>|z|     [95% conf. interval]
------------------------------------------------------------------------------
           W.y    0.302192    0.015724    19.22   0.000    0.271373    0.333010
          L1.y    0.390505    0.008521    45.83   0.000    0.373803    0.407206
            x1    1.001188    0.009817   101.98   0.000    0.981946    1.020430
            x2    0.506094    0.010803    46.85   0.000    0.484920    0.527268
------------------------------------------------------------------------------
       sigma_f    1.404462  (std. dev. of factor error component)
       sigma_e    0.964170  (std. dev. of idiosyncratic error component)
           rho    0.679676  (fraction of variance due to factors)
------------------------------------------------------------------------------
Hansen test of the overidentifying restrictions  chi2(4) =    2.9647
H0: overidentifying restrictions are valid        Prob > chi2 =    0.5638
```

The eigenvalue-ratio criterion selects the true number of factors in each instrument block and in the residuals, the estimates are close to the true values, and the J test does not reject. Ignoring the factors (`factmax=0`) yields IV estimates based on raw covariates:

```
>>> res0 = sd.fit(data, "y", ["x1", "x2"], "id", "time", W, tlags=1,
...               iv_lags=1, factmax=0)

        2SIV  no factors
W.y   0.3022      0.3304
L1.y  0.3905      0.3839
x1    1.0012      1.3825
x2    0.5061      0.8405
J p-values: 2SIV 0.564, no factors 0.0231
```

The slope estimates are biased upwards by 0.34 to 0.38 because the covariates are correlated with the omitted factor component, and the J test rejects at the 5% level.

Impacts. Long-run impacts with simulation-based inference are requested with

```
>>> lr_delta = res.impacts("lr", method="delta")
>>> lr_sim = res.impacts("lr", method="simulation", draws=5000, seed=1)
>>> print(lr_sim.round(4))
>>> print(pd.concat({"delta": lr_delta[["se", "lower", "upper"]],
...                  "simulation": lr_sim[["se", "lower", "upper"]]},
...                 axis=1).loc["total"].round(4))
```

```
                  impact      se        z    p   lower   upper
effect   variable
direct   x1        1.7250  0.0272  63.4255  0.0  1.6740  1.7805
         x2        0.8720  0.0220  39.7039  0.0  0.8284  0.9144
indirect x1        1.5330  0.1445  10.6091  0.0  1.2697  1.8312
         x2        0.7749  0.0733  10.5786  0.0  0.6400  0.9295
total    x1        3.2580  0.1576  20.6760  0.0  2.9698  3.5840
         x2        1.6469  0.0835  19.7285  0.0  1.4953  1.8242
           delta                 simulation
              se   lower   upper         se   lower   upper
variable
x1        0.1595  2.9454  3.5706     0.1576  2.9698  3.5840
x2        0.0849  1.4805  1.8133     0.0835  1.4953  1.8242
```

The second panel compares the two methods for the total impacts. The simulation intervals are shifted to the right of the symmetric Delta intervals, reflecting the convexity of $\beta/(1-\rho-\psi)$ in the persistence parameters. The true long-run total impacts are $1/0.3 = 3.333$ for `x1` and 1.667 for `x2`; both intervals cover them.

Safeguards. Passing the weights as an unlabelled array is refused, and reversing the order of a labelled matrix leaves the estimates unchanged:

```
>>> sd.fit(data, "y", ["x1", "x2"], "id", "time", W.to_numpy())
>>> perm = W.index[::-1]
>>> res_p = sd.fit(data, "y", ["x1", "x2"], "id", "time", W.loc[perm, perm],
...                tlags=1, iv_lags=1)
```

```
WeightsAlignmentError: An unlabelled weights array was supplied. Pass a
labelled DataFrame, or the unit labels via `W_ids`, or set
`assume_ordered=True` to accept positional matching.
max |difference|: 1.1102230246251565e-15
```

Heterogeneous slopes. The next data set has heterogeneous slopes, centred on the same values, and 3% of units (nine units) whose covariate slopes are shifted by +3, mimicking units with data problems or a different structure.

```
>>> dh, Wh, th = sd.simulate_panel(N=300, T=40, het=0.2, outlier_share=0.03,
...                                seed=7)
>>> mg = sd.fit(dh, "y", ["x1", "x2"], "id", "time", Wh, tlags=1,
...             iv_lags=1, estimator="mg")
>>> print(mg.summary())
```

```
Defactored instrumental variables estimation
Number of obs               =      11700   Number of units     =      300
Number of instruments       =          8   Obs per unit (min/avg/max)= 39/39.0/39
Factors in instruments      =        2/2   Factors in u        =        -
Mean-group estimator (model with heterogeneous slope coefficients)
------------------------------------------------------------------------------
                    Coef.   Std. err.        z   P>|z|     [95% conf. interval]
------------------------------------------------------------------------------
          W.y    0.281662    0.018684    15.08   0.000    0.245043    0.318281
         L1.y    0.389689    0.009815    39.70   0.000    0.370452    0.408925
           x1    1.071706    0.036534    29.33   0.000    1.000101    1.143311
           x2    0.597771    0.034591    17.28   0.000    0.529974    0.665568
------------------------------------------------------------------------------
```

The MG estimates of the covariate slopes, 1.072 and 0.598, lie well above the centre of the regular units, 1 and 0.5. The influence diagnostics show why:

```
>>> infl = sd.mg_influence(mg, threshold=3.5)
>>> cols = ["unit", "parameter", "theta_i", "mod_z", "dfmg", "flag"]
>>> print(infl.sort_values("dfmg", key=abs, ascending=False)[cols].head(6))
>>> flagged = sd.flagged_units(infl, parameters=["x1", "x2"])
```

```
      unit parameter  theta_i    mod_z    dfmg  flag
375     76      L1.y   1.5416   8.0482  0.3925  True
255    256       W.y  -1.7167  -7.6234 -0.3577  True
1074   175        x2   3.9455  10.1488  0.3237  True
1082   183        x2   3.9051  10.0291  0.3198  True
810    211        x1   4.4822   9.9748  0.3122  True
1111   212        x2   3.7138   9.4627  0.3013  True
flagged: [31, 111, 114, 175, 183, 211, 212, 223, 294, 300]
true aberrant units: [31, 111, 175, 183, 211, 212, 223, 294, 300]
```

Each of the most influential units moves an MG coefficient by about a third of its standard error. Flagging on the covariate slopes recovers all nine aberrant units and one regular unit. The two units flagged for the spatial and time lag coefficients have imprecise unit-level estimates; with $T = 40$ per-unit IV estimates of these coefficients are noisy, which is why flagging is best restricted to the parameters of interest. Robust summaries and the MG estimate without the flagged units are

```
>>> print(sd.mg_robust(mg, method="trimmed", trim=0.1, bootstrap=999, seed=1))
>>> print(sd.mg_robust(mg, method="mean", exclude=flagged, bootstrap=999,
...                    seed=1))
```

```
          MG   MG_se  trimmed      se   lower   upper
W.y   0.2817  0.0187   0.2843  0.0157  0.2508  0.3142
L1.y  0.3897  0.0098   0.3929  0.0090  0.3759  0.4125
x1    1.0717  0.0365   1.0000  0.0209  0.9595  1.0397
x2    0.5978  0.0346   0.5200  0.0198  0.4828  0.5592
          MG   MG_se    mean      se   lower   upper
W.y   0.2780  0.0192  0.2780  0.0194  0.2419  0.3161
L1.y  0.3881  0.0100  0.3881  0.0097  0.3691  0.4072
x1    0.9839  0.0196  0.9839  0.0192  0.9454  1.0206
x2    0.5110  0.0185  0.5110  0.0183  0.4744  0.5453
```

Both the trimmed mean and the MG estimate without the flagged units are close to the centre of the regular units, with standard errors about half those of the full MG estimate.

Staggered entry. Finally, 30% of the units enter the panel at random dates during the first half of the sample.

```
>>> du, Wu, tu = sd.simulate_panel(N=300, T=40, entry_share=0.3, seed=11)
>>> ru = sd.fit(du, "y", ["x1", "x2"], "id", "time", Wu, tlags=1,
...             iv_lags=1, unbalanced="staggered")

time
1    210
2      4
3      3
4      5
Name: count, dtype: int64
UserWarning: Estimation with an unbalanced panel is experimental: the asymptotic theory of Cui et a
Defactored instrumental variables estimation
Number of obs               =      10800   Number of units     =      300
Number of instruments       =          8   Obs per unit (min/avg/max)= 20/36.0/39
Factors in instruments      =        2/2   Factors in u        =        2
Unbalanced panel (staggered); experimental.
Second-stage estimator (model with homogeneous slope coefficients)
------------------------------------------------------------------------------
                     Coef.   Std. err.        z   P>|z|     [95% conf. interval]
------------------------------------------------------------------------------
           W.y    0.316191    0.012070    26.20   0.000    0.292533    0.339848
          L1.y    0.406260    0.008640    47.02   0.000    0.389327    0.423194
            x1    0.994829    0.011469    86.74   0.000    0.972349    1.017308
            x2    0.517562    0.010480    49.39   0.000    0.497022    0.538102
------------------------------------------------------------------------------
       sigma_f    1.245863  (std. dev. of factor error component)
       sigma_e    0.961159  (std. dev. of idiosyncratic error component)
           rho    0.626888  (fraction of variance due to factors)
------------------------------------------------------------------------------
Hansen test of the overidentifying restrictions  chi2(4) =    4.4186
H0: overidentifying restrictions are valid        Prob > chi2 =    0.3523
```

The first lines show how many units start in each of the first periods. The estimator uses all 10,800 available observations and issues a warning that the option is experimental.

		ψ_0			ρ_1			β_1			
(N, T)	Estimator	Bias	RMSE	Size	Bias	RMSE	Size	Bias	RMSE	Size	J
(50, 50)	2SIV	0.005	0.034	9.1	−0.003	0.020	11.1	0.002	0.026	10.4	4.9
	First-stage IV	0.021	0.050	12.3	−0.018	0.030	8.1	0.012	0.029	7.5	18.0
	IV, no factors	0.028	0.036	41.6	−0.010	0.016	30.4	0.354	0.357	100.0	21.5
(100, 100)	2SIV	0.001	0.015	6.5	−0.001	0.009	9.6	0.000	0.012	7.6	3.5
	First-stage IV	0.003	0.018	6.6	−0.009	0.016	7.5	0.005	0.014	6.5	14.8
	IV, no factors	0.019	0.023	45.6	−0.007	0.010	40.6	0.360	0.361	100.0	54.0
(200, 200)	2SIV	0.000	0.007	5.8	0.000	0.004	5.3	0.000	0.006	4.5	3.8
	First-stage IV	0.001	0.008	6.6	−0.004	0.007	5.5	0.003	0.007	6.2	17.4
	IV, no factors	0.012	0.015	49.2	−0.004	0.007	46.1	0.363	0.363	100.0	91.1

Table 2: Homogeneous slopes: bias, root mean squared error, and size (%) of 5% t tests for $\psi_0 = 0.3$, $\rho_1 = 0.4$, $\beta_1 = 1$, and rejection frequency (%) of the J test. 1,000 replications.

5. Monte Carlo evidence

All experiments use the data-generating process of Section 4, 1,000 replications, a new weights matrix and new loadings in each replication, and the specification with one time lag, one instrument lag, and spatial lags of the instruments (eight instruments, four parameters). The code is in `replication/monte_carlo.py`.

5.1. Homogeneous slopes

Table 2 compares the 2SIV estimator with the first-stage IV estimator and with IV estimation that ignores the factors (`factmax=0`), for $(N, T) \in \{(50, 50), (100, 100), (200, 200)\}$. Size refers to two-sided t tests at the 5% level, and the last column reports the rejection frequency of the J test at the 5% level.

The 2SIV estimator is practically unbiased in all designs, and its RMSE halves as N and T double. Its t tests over-reject somewhat in the smallest design (sizes between 9.1% and 11.1%), but their size is between 4.5% and 5.8% when $N = T = 200$, and the J test has size between 3.5% and 4.9%. The eigenvalue-ratio criterion selects the true number of factors in the instruments in at least 99.8% of the replications and in the residuals in at least 97.2%. The first-stage estimator is consistent but less precise; its J test rejects in 15%–18% of the replications because it ignores the factors in the error, which illustrates why the second stage is the default. Ignoring the factors produces a bias of about 0.36 in β_1 that does not vanish as the sample grows, so that t tests on β_1 always reject. Reassuringly, the J test detects this misspecification with a frequency that rises from 21.5% to 91.1%.

5.2. Inference on long-run impacts

To make the long-run multipliers strongly non-linear, Table 3 uses $\psi_0 = 0.35$ and $\rho_1 = 0.5$, so that $1 - \rho_1 - \psi_0 = 0.15$ and the total long-run impact of x_1 is 6.67. It reports the coverage of nominal 95% intervals, the frequencies with which the true value lies below or above the interval, and the median interval width. In the smallest design, the point estimates violated the stability condition in 22 of the 1,000 replications, for which impacts are undefined; these replications are excluded.

The results give a nuanced picture that users should know about. The Delta intervals are

		Delta method				Simulation (Krinsky–Robb)			
(N, T)	Impact	Cover	Below	Above	Width	Cover	Below	Above	Width
(30, 30)	direct	88.7	1.7	9.6	0.63	87.2	7.1	5.7	0.71
	indirect	89.1	0.2	10.7	5.60	86.9	7.6	5.5	7.15
	total	88.7	0.2	11.1	6.05	87.1	7.5	5.4	7.60
(50, 50)	direct	92.7	1.6	5.7	0.40	90.4	5.6	4.0	0.41
	indirect	90.8	0.5	8.7	3.76	88.6	6.4	5.0	4.15
	total	90.8	0.5	8.7	4.08	88.3	6.4	5.3	4.48
(100, 100)	direct	92.9	2.2	4.9	0.19	93.7	3.2	3.1	0.19
	indirect	93.3	1.4	5.3	1.85	94.5	2.7	2.8	1.89
	total	93.5	1.4	5.1	2.00	94.2	2.6	3.2	2.04

Table 3: Long-run impacts of x_1 with $\psi_0 = 0.35$, $\rho_1 = 0.5$: coverage (%) of nominal 95% intervals, frequency (%) with which the true value lies below the lower or above the upper limit, and median width.

symmetric while the sampling distribution of the impacts is skewed to the right; as a result almost all of their non-coverage is on one side: the true value lies above the upper limit in 9%–11% of the replications for indirect and total impacts when $N, T \leq 50$, and below the lower limit in less than 1%. Simulation intervals spread the errors almost evenly between the two tails. Their overall coverage, however, is not better in small samples (86.9%–88.6% compared with 88.7%–90.8% for the Delta method), because both methods inherit the same first-order approximation of the variance. In the largest design both methods are close to nominal coverage, simulation slightly more so (94.2% compared with 93.5% for total impacts). The practical recommendation is therefore to report simulation intervals when the process is persistent, because they convey the asymmetry of the uncertainty, while recognizing that neither method is exact in small samples.

5.3. Mean-group estimation with aberrant units

The data for Table 4 have heterogeneous slopes, with ψ_{0i} and ρ_{1i} uniform on ± 0.1 around their means and $\beta_{\ell i}$ uniform on ± 0.4 around β_ℓ. A share of 0%, 5%, or 10% of the units is *aberrant*: both of their covariate slopes are shifted by $+3$. The target is the mean of the regular units, $\beta_1 = 1$. The table reports the MG estimator, the coefficient-wise 10% trimmed mean and the median of the unit-specific estimates (with 499 bootstrap replications over units), and the MG estimator after removing the units flagged on β_1 or β_2, together with the true and false positive rates of the flags.

Without aberrant units all four estimators are centred on the truth and have coverage between 94.2% and 95.3%; the median pays an efficiency price (RMSE 0.041 against 0.031 for MG when $N = 100$), and the flags mark at most 0.3% of the regular units. With 5% aberrant units the MG estimator is biased by 0.15, and its coverage falls to 36% for $N = 100$ and to zero for $N = 200$: a larger sample only makes the wrong answer more precise. The trimmed mean and the median remove most of the bias but not all of it, because the symmetric trimming also cuts regular units from the lower tail, so their coverage remains below nominal (82%–92%) and deteriorates when 10% of the units are aberrant. Excluding the flagged units restores both the centre and the coverage (93%–95%) in all designs.

With a shift of $+3$ the aberrant units are far outside the distribution of the regular slopes, which makes detection easy: every aberrant unit is flagged in every design. To see how the

		No aberrant units			5% aberrant			10% aberrant		
(N, T)	Estimator	Bias	RMSE	Cover	Bias	RMSE	Cover	Bias	RMSE	Cover
(100, 50)	MG	0.001	0.031	94.5	0.152	0.155	36.3	0.300	0.302	0.0
	Trimmed MG (10%)	0.002	0.031	95.1	0.031	0.044	87.8	0.065	0.073	63.7
	Median	0.001	0.041	95.3	0.026	0.050	91.7	0.049	0.066	82.0
	MG excl. flagged	0.002	0.031	94.3	0.003	0.032	95.2	0.000	0.034	93.1
	Flagging: TPR / FPR		– / 0.3			100.0 / 0.2			100.0 / 0.1	
(200, 100)	MG	0.000	0.019	94.2	0.150	0.151	0.0	0.300	0.300	0.0
	Trimmed MG (10%)	0.000	0.021	94.3	0.023	0.031	81.6	0.054	0.058	40.4
	Median	−0.001	0.029	94.8	0.020	0.035	89.9	0.044	0.054	70.4
	MG excl. flagged	0.000	0.019	94.2	0.000	0.019	94.0	0.000	0.020	94.4
	Flagging: TPR / FPR		– / 0.0			100.0 / 0.0			100.0 / 0.0	

Table 4: Heterogeneous slopes with aberrant units (slopes shifted by +3): bias, RMSE, and coverage (%) of nominal 95% intervals for the centre of the regular units, $\beta_1 = 1$, and true / false positive rates (%) of the outlier flags. 1,000 replications.

tools behave when detection is hard, we repeated the $(N, T) = (200, 100)$ design with a shift of +1, only 2.5 times the half-width of the distribution of regular slopes (1,000 replications; `monte_carlo.py`, experiment `3b`). The flags then identify 55.5% of the aberrant units when 5% are aberrant and 39.6% when 10% are, with no false positives. Excluding the flagged units still halves the bias of the MG estimator with 5% aberrant units (0.020 against 0.050) but no longer restores coverage (84.6%), and with 10% aberrant units it is no better than the trimmed mean (bias 0.056 and 0.053). The median has the smallest bias in both cases (0.022 and 0.044). The diagnostics are therefore most useful when aberrant units are clearly separated, which is the case they are designed to reveal; when contamination blends into the regular distribution, robust summaries and exclusion give similar, partially corrected answers, and the gap between them and the MG estimate is itself the informative signal.

5.4. Unbalanced panels

Table 5 considers $N = T = 100$ with two kinds of missing data. With *staggered entry*, 30% of the units enter at a random date during the first half of the sample and, consistently with Section 2.6, do not interact with other units before they enter. The experimental unbalanced estimator is compared with the two balanced alternatives a user of existing software would have: dropping the late units (and re-standardizing the weights matrix of the remaining units) or keeping all units but only the periods in which all of them are observed. With *sporadic* gaps, 2% of the unit-period cells are missing at random; the only balanced alternative is to keep the units without gaps.

Under staggered entry the unbalanced estimator is centred on the true values, and its sizes (6.6%–8.6%) are close to those of the 2SIV estimator in balanced panels of the same dimensions in Table 2. Dropping the late units is not a harmless loss of information: it removes nodes from the network and biases the spatial coefficient by −0.07, so that t tests on ψ_0 reject in 98% of the replications. Restricting the sample to the common period is valid but discards almost half of the observations, which raises the RMSE of ψ_0 from 0.015 to 0.023. With 2% sporadic gaps the unbalanced estimator is again nearly unbiased (bias of ψ_0 equal to −0.003), whereas keeping only the units without gaps leaves about 14% of the observations and severely biases the spatial coefficient. These results support the usefulness of the option in designs like these.

			ψ_0			ρ_1			β_1		
Design	Estimator	NT	Bias	RMSE	Size	Bias	RMSE	Size	Bias	RMSE	Size
Staggered	Unbalanced, all units	9149	0.001	0.015	6.6	0.001	0.009	8.1	0.003	0.013	8.6
	Drop late units	6930	−0.070	0.073	97.6	0.003	0.012	10.0	0.004	0.016	9.5
	Common period	5112	0.002	0.023	7.9	−0.002	0.013	8.9	0.002	0.018	8.2
Sporadic	Unbalanced, all units	9507	−0.003	0.016	7.4	0.000	0.009	9.3	0.001	0.012	7.5
	Complete units only	1307	−0.222	0.231	97.1	0.009	0.037	25.2	0.019	0.055	26.8

Table 5: Unbalanced panels, $N = T = 100$: average number of observations used, bias, RMSE, and size (%) of 5% t tests. 1,000 replications.

They do not replace a formal theory, and the sporadic case in particular could behave worse when missingness is more frequent or related to the outcome; the option therefore remains labelled experimental.

6. Empirical application: credit risk in US banks

This section uses the data analysed by Kripfganz and Sarafidis (2025): 350 US banks observed over 36 quarters (2006Q1–2014Q4), distributed with the replication materials of their article. The dependent variable is the ratio of non-performing loans to total loans (`NPL`), a measure of credit risk. Operational inefficiency (`INEFF`) is treated as endogenous with respect to the idiosyncratic error and instrumented by the ratio of interest expenses to deposits (`INTEREST`); capital adequacy (`CAR`), size (`SIZE`), the capital buffer (`BUFFER`), profitability (`PROFIT`), asset quality (`QUALITY`) and liquidity (`LIQUIDITY`) are exogenous with respect to the idiosyncratic error but may be correlated with the factors. The row-standardized weights matrix links each bank to the banks whose debt ratios are most strongly correlated with its own. The weights file is an unlabelled matrix ordered like the bank identifiers, so it is passed together with these identifiers:

```
>>> banks = pd.read_stata("v113i06.dta")
>>> W = pd.read_csv("W.csv", header=None).to_numpy()
>>> ids = sorted(banks["ID"].unique())
>>> common = ["CAR", "SIZE", "BUFFER", "PROFIT", "QUALITY", "LIQUIDITY"]
>>> X, IV = ["INEFF"] + common, ["INTEREST"] + common
>>> spec = dict(W_ids=ids, splag=True, tlags=1, iv=IV, iv_lags=1,
...             iv_splags=True, std=True)
>>> res = sd.fit(banks, "NPL", X, "ID", "TIME", W, **spec)
>>> mg = sd.fit(banks, "NPL", X, "ID", "TIME", W, estimator="mg", **spec)
```

The specification has 28 instruments for 9 parameters. All results of this section are produced by `replication/bank_replication.py`.

6.1. Cross-software check

Table 6 compares the estimates with those printed in Tables 1 and 5 of Kripfganz and Sarafidis (2025). For both the 2SIV and the mean-group estimator, the coefficients and standard errors agree with the published ones in every printed digit: the largest absolute difference is

	2SIV (Table 1)				Mean group (Table 5)			
	Coefficient		Std. error		Coefficient		Std. error	
	spdfiv	Stata	**spdfiv**	Stata	**spdfiv**	Stata	**spdfiv**	Stata
`W.NPL`	0.3943	0.3943	0.0849	0.0849	0.0316	0.0316	0.0511	0.0511
`L1.NPL`	0.2899	0.2899	0.0544	0.0544	0.3005	0.3005	0.0149	0.0149
`INEFF`	0.4474	0.4474	0.1046	0.1046	0.7588	0.7588	0.1584	0.1584
`CAR`	0.0305	0.0305	0.0058	0.0058	0.2181	0.2181	0.0263	0.0263
`SIZE`	0.2226	0.2226	0.0942	0.0942	2.0040	2.0040	0.3385	0.3385
`BUFFER`	−0.0545	−0.0545	0.0119	0.0119	−0.3764	−0.3764	0.0420	0.0420
`PROFIT`	−0.0053	−0.0053	0.0018	0.0018	−0.0180	−0.0180	0.0059	0.0059
`QUALITY`	0.1830	0.1830	0.0308	0.0308	0.2873	0.2873	0.1387	0.1387
`LIQUIDITY`	2.4524	2.4524	0.2696	0.2696	6.3302	6.3302	0.5059	0.5059
J statistic	18.825	18.825						
σ_f	0.6416	0.6416						
σ_e	0.9038	0.9038						

Table 6: Bank data: estimates from **spdfiv** and from `spxtivdfreg` as published in Tables 1 and 5 of Kripfganz and Sarafidis (2025).

below 10^{-6}, the precision of the printed numbers. The J statistic (18.825), the factor and idiosyncratic standard deviations, the numbers of factors, and the long-run direct, indirect and total impacts of Table 3 of the original article are reproduced to the same precision. A test in the package (`tests/test_stata_replication.py`) checks this agreement whenever the replication materials are available.

Reaching this agreement required five implementation details of `spxtivdfreg` that are not stated in the equations of the article and that we identified from the source code of the command, distributed under the GPL with the article. (i) With option `std`, variables are standardized across banks *within each period* before the factors are extracted. (ii) The eigenvalue-ratio criterion chooses between one and `factmax` factors, without the mock eigenvalue of Ahn and Horenstein (2013). (iii) All lag blocks of a set of instruments use the largest number of factors selected for any of them. (iv) The variance of the 2SIV estimator and the J statistic use the weight matrix computed from the first-stage residuals, as in Equation (11) of the article. (v) The instruments are demeaned before, but not after, the factors are projected out. Point (v) leaves $\hat{\boldsymbol{Z}}_i^\top \boldsymbol{C}_i$ and $\hat{\boldsymbol{Z}}_i^\top \boldsymbol{y}_i$ unchanged, because the regressors are demeaned, but it changes the IV weight $\hat{\boldsymbol{Z}}_i^\top \hat{\boldsymbol{Z}}_i$; its effect on the pooled estimator is of the order 10^{-4}, whereas for the mean-group estimator, which uses a separate weight for each bank, it changes the spatial coefficient from 0.087 to 0.032. Points (i)–(v) are the defaults of **spdfiv**; points (i), (ii) and (iv) can be changed through arguments of `fit()` (Table 1). Without them, the selected numbers of factors differ (one instead of two in the instruments, none instead of one in the residuals), and the coefficient on `LIQUIDITY` is 1.50 instead of 2.45. Making such details explicit is, in our view, a useful by-product of an independent implementation: they matter for the results but cannot be inferred from the published equations.

6.2. Long-run impacts

Table 7 reports the long-run impacts with Delta-method and simulation intervals (10,000 draws, all of which satisfy the stability condition). The two methods agree for the direct impacts: the simulation intervals are on average as wide as the Delta intervals, with a median width ratio of 1.01. They disagree for the indirect impacts, which depend on the multiplier

			Delta method		Simulation	
Impact	Variable	Estimate	s.e.	95% interval	s.e.	95% interval
direct	INEFF	0.647	0.159	[0.335, 0.959]	0.163	[0.339, 0.979]
	CAR	0.044	0.009	[0.026, 0.062]	0.009	[0.027, 0.063]
	SIZE	0.322	0.142	[0.044, 0.600]	0.143	[0.056, 0.614]
	BUFFER	−0.079	0.018	[−0.115, −0.043]	0.018	[−0.117, −0.044]
	PROFIT	−0.008	0.002	[−0.012, −0.003]	0.002	[−0.012, −0.003]
	QUALITY	0.265	0.047	[0.173, 0.356]	0.047	[0.179, 0.363]
	LIQUIDITY	3.547	0.445	[2.674, 4.420]	0.462	[2.736, 4.513]
indirect	INEFF	0.769	0.335	[0.112, 1.427]	0.588	[0.294, 1.909]
	CAR	0.052	0.024	[0.006, 0.099]	0.042	[0.020, 0.138]
	SIZE	0.383	0.198	[−0.004, 0.770]	0.285	[0.071, 1.043]
	BUFFER	−0.094	0.043	[−0.178, −0.010]	0.074	[−0.244, −0.035]
	PROFIT	−0.009	0.005	[−0.018, 0.000]	0.007	[−0.024, −0.002]
	QUALITY	0.315	0.141	[0.039, 0.591]	0.244	[0.121, 0.820]
	LIQUIDITY	4.218	1.742	[0.803, 7.633]	3.100	[1.801, 10.548]
total	INEFF	1.417	0.427	[0.579, 2.254]	0.659	[0.724, 2.731]
	CAR	0.097	0.029	[0.039, 0.154]	0.046	[0.052, 0.191]
	SIZE	0.705	0.310	[0.097, 1.312]	0.380	[0.140, 1.529]
	BUFFER	−0.173	0.054	[−0.279, −0.066]	0.083	[−0.343, −0.089]
	PROFIT	−0.017	0.006	[−0.029, −0.004]	0.008	[−0.034, −0.006]
	QUALITY	0.580	0.167	[0.252, 0.907]	0.265	[0.334, 1.136]
	LIQUIDITY	7.765	1.904	[4.034, 11.496]	3.262	[5.021, 14.366]

Table 7: Bank data: long-run average impacts with 95% intervals from the Delta method and from Krinsky–Robb simulation (10,000 draws).

$1/(1-\rho_1-\psi_0)$ with $\hat{\rho}_1+\hat{\psi}_0=0.68$. The simulation intervals are skewed to the right, and are 1.25 times as wide as the Delta intervals in the median. The disagreement changes a conclusion: the Delta interval for the indirect impact of SIZE, [−0.004, 0.770], contains zero, whereas the simulation interval, [0.071, 1.043], does not. Given the Monte Carlo evidence of Section 5, we read this as showing that the uncertainty about the spillovers is asymmetric, not as establishing that the simulation interval is the more accurate one in this sample.

6.3. What drives the mean-group estimates?

Kripfganz and Sarafidis (2025) found that the MG estimates of some slopes differ from the 2SIV estimates by an order of magnitude, although the J test does not reject the homogeneous model, and that the spatial spillover disappears under heterogeneous slopes; they suggested that a few outlying banks might be responsible. The diagnostics of Section 2.5 allow this conjecture to be examined.

Table 8 shows that single banks move the MG estimate of SIZE or BUFFER by about 0.19 to 0.27 standard errors. Their instrument matrices are far from singular (condition numbers below 10^8), so these are not numerical artefacts. The flags mark 24 banks for SIZE, 15 for BUFFER, and 95 of the 350 banks for at least one covariate slope. The distribution of the bank-specific estimates is therefore heavy tailed rather than contaminated by a handful of outliers.

Table 9 gives the robust summaries. For SIZE the median, 0.807, closes 67% of the gap between the MG estimate (2.004) and the 2SIV estimate (0.223); for BUFFER the median closes 49% of the gap (−0.220 against −0.376 and −0.055). Excluding the flagged banks gives similar values (0.707 and −0.234). The spatial coefficient behaves differently: the median (0.034) and the MG estimate without the flagged banks (0.023) are as small as the MG estimate itself

Bank	Coefficient	$\hat{\theta}_i$	Modified z	Change in MG (s.e.)	cond($\boldsymbol{Z}_i^\top \boldsymbol{Z}_i$)
147	BUFFER	3.618	7.58	0.272	1.7×10^6
26	SIZE	32.324	9.14	0.257	1.7×10^6
112	SIZE	30.991	8.76	0.245	3.7×10^6
296	SIZE	30.810	8.70	0.244	2.5×10^7
276	BUFFER	2.726	5.82	0.212	1.0×10^7
199	BUFFER	−3.368	−6.22	−0.204	1.9×10^7
79	SIZE	25.680	7.22	0.200	8.6×10^5
16	BUFFER	−3.207	−5.90	−0.193	1.2×10^7

Table 8: Bank data: the eight banks with the largest influence on the MG estimates of the SIZE and BUFFER slopes. "Change in MG" is the change in the MG estimate from deleting the bank, in units of its standard error.

	MG	Trimmed (10%)	Median	MG excl. flagged
W.NPL	0.032	0.020	0.034	0.023
	[−0.069, 0.132]	[−0.063, 0.102]	[−0.031, 0.117]	[−0.065, 0.109]
L1.NPL	0.301	0.301	0.301	0.303
	[0.271, 0.330]	[0.272, 0.334]	[0.268, 0.345]	[0.271, 0.334]
INEFF	0.759	0.507	0.313	0.191
	[0.448, 1.069]	[0.271, 0.763]	[0.099, 0.572]	[−0.001, 0.403]
CAR	0.218	0.184	0.123	0.133
	[0.167, 0.270]	[0.143, 0.225]	[0.094, 0.156]	[0.094, 0.171]
SIZE	2.004	1.282	0.807	0.707
	[1.341, 2.668]	[0.796, 1.750]	[0.232, 1.198]	[0.300, 1.123]
BUFFER	−0.376	−0.321	−0.220	−0.234
	[−0.459, −0.294]	[−0.389, −0.254]	[−0.281, −0.167]	[−0.301, −0.171]
PROFIT	−0.018	−0.016	−0.008	−0.020
	[−0.030, −0.006]	[−0.025, −0.007]	[−0.018, −0.001]	[−0.030, −0.011]
QUALITY	0.287	0.179	0.081	0.159
	[0.015, 0.559]	[0.021, 0.331]	[−0.055, 0.214]	[0.011, 0.304]
LIQUIDITY	6.330	4.921	3.200	3.383
	[5.339, 7.322]	[4.075, 5.772]	[2.573, 4.049]	[2.691, 4.033]
Units	350	350	350	255

Table 9: Bank data: MG estimates and robust summaries of the bank-specific estimates, with 95% intervals (normal for MG, bootstrap percentile with 999 resamples of banks for the others).

(0.032) and far below the 2SIV estimate (0.394). Five banks have nearly singular instrument matrices, because QUALITY does not vary over time for them, but removing them leaves the MG estimates practically unchanged (SIZE: 1.997).

The conjecture is thus partly supported. Extreme bank-level estimates account for about two thirds of the difference in the SIZE slope and for about half of it in the BUFFER slope, but not for the disappearance of the spatial spillover, which is a feature of the bulk of the distribution. The MG estimate of the spatial coefficient is also sensitive to the construction of the instruments: without double defactoring it is 0.151 instead of 0.032, and point (v) above moves it as much. One explanation consistent with this evidence, which we do not pursue here, is the finite-sample behaviour of bank-level IV estimation with 28 instruments and 35 observations per bank, a ratio at which IV estimates are known to be pulled towards least squares.

	Full panel		Staggered, all banks		Late banks dropped	
	Coef.	s.e.	Coef.	s.e.	Coef.	s.e.
W.NPL	0.394	0.085	0.352	0.106	0.367	0.105
L1.NPL	0.290	0.054	0.305	0.054	0.243	0.058
INEFF	0.447	0.105	0.435	0.101	0.456	0.117
CAR	0.031	0.006	0.032	0.006	0.030	0.006
SIZE	0.223	0.094	0.199	0.105	0.330	0.124
BUFFER	−0.055	0.012	−0.051	0.013	−0.055	0.013
PROFIT	−0.005	0.002	−0.004	0.002	−0.006	0.002
QUALITY	0.183	0.031	0.173	0.031	0.182	0.029
LIQUIDITY	2.452	0.270	2.525	0.282	2.468	0.325
Observations	12,250		11,602		8,575	
Banks	350		350		245	

Table 10: Bank data with artificial staggered entry: 2SIV estimates.

N	T	2SIV	Impacts (1000 draws)	MGIV
100	50	0.03	0.02	0.04
250	50	0.06	0.06	0.09
500	50	0.11	0.26	0.19
1000	50	0.33	1.57	0.48
100	200	0.05	0.01	0.06
500	200	0.23	0.25	0.32

Table 11: Computing time in seconds (best of three runs).

6.4. Staggered entry

Finally, the bank panel is made unbalanced artificially, so that the answer is known: for a random 105 of the 350 banks the first 1 to 12 quarters are deleted. Table 10 compares three estimates: the full balanced panel, the experimental unbalanced estimator using all 11,602 remaining observations, and the balanced alternative that drops the late banks (8,575 observations).

The unbalanced estimates stay within 0.58 standard errors of the full-panel estimates. Dropping the late banks moves the estimates by up to 1.14 standard errors, because it discards 30% of the network as well as the observations. In this application the experimental option thus recovers more of the full-sample information than the balanced alternative available in existing software.

7. Computing time

Table 11 reports the best of three wall-clock times, in seconds, for the 2SIV and MGIV estimators and for long-run impacts with 1,000 simulation draws, on the platform described in the computational details. Estimation with a thousand units and fifty periods takes less than half a second; the impact computation is dominated by the Schur decomposition of $\boldsymbol{W}$, which is computed once.

8. Summary and discussion

spdfiv makes the defactored IV approach to spatial dynamic panels with interactive effects available as open-source software. It reproduces the defining equations of the 2SIV estimator to machine precision. On the data of Kripfganz and Sarafidis (2025) it reproduces the published 2SIV and mean-group results of `spxtivdfreg` in every printed digit, and the comparison made explicit five implementation details of that command. The package adds safeguards and diagnostics that matter in applied work: label-based alignment of the weights matrix, exact and fast computation of impacts with a choice between Delta-method and simulation inference, tools that show which units drive mean-group estimates, and an experimental treatment of unbalanced panels.

Several limitations remain. The asymptotic theory for unbalanced panels and for the robust MG summaries has not been developed; the corresponding options are diagnostics and are labelled as such. The package does not implement a test of remaining cross-sectional dependence, because the standard CD test is invalid when factors have been estimated from the same data (Juodis and Reese 2022). Planned extensions include an interface for **PySAL** weights objects, a test of slope homogeneity, and an R port.

Computational details

The results in this paper were obtained using Python 3.13.17 with **spdfiv** 0.1.0, **NumPy** 2.5.3, **SciPy** 1.18.1, and **pandas** 3.0.5, on Linux with two CPU cores. The package and the replication materials are available from `https://github.com/abdulameert-boop/spdfiv`.

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